

CPP SAT Math Concept Reference

College Prep Program • Student Reference

A complete resource covering every tested SAT Math skill: recognize the question type, apply a repeatable method, and review with Khan Academy.



Download this reference

Scan the code, or visit:

khalascholars.org/psat-math

How to Use This Reference

- Translate the problem into mathematical relationships before calculating.
- Write units and define variables; this prevents many setup errors.
- Choose the fastest valid method: algebra, graphing, substitution, elimination, or estimation.
- Use Desmos to verify—not replace—the mathematical setup.
- After a missed question, review the linked Khan Academy lesson and solve a similar problem without notes.
- After each Bluebook test, look up every skill from your error log here, then complete the matching Khan Academy practice.

Quick-Reference Map

Use this page to identify the skill quickly. Turn to the full section for the method, traps, example, and self-check questions. The Khan Academy references under each section link directly to that skill's lesson article in the SAT Math course — click the lesson name to open it.

Skill	Recognize it	First move
Linear Equations (One Variable)	One variable, no exponents	Clear parentheses and fractions first.
Linear Functions & Graphs	Constant rate; slope or “per” language	Interpret slope as a rate with units.
Systems of Linear Equations	Two equations share variables	Choose substitution or elimination.
Linear Inequalities	At least / at most / range of values	Write each constraint separately.
Equivalent Expressions	“Which expression is equivalent”	Scan choices for the target form.
Isolating Quantities	Solve a formula for one letter	Undo operations in reverse order.
Quadratic Equations	x^2 term; find roots or solutions	Factor quickly or use the formula.
Linear–Quadratic Systems	Line and parabola together	Substitute to create one quadratic.
Radical / Rational / Abs. Value	$\sqrt{\quad}$, variable denominator, or $ x $	Isolate it, solve, check every root.
Nonlinear Functions	Evaluate, compose, or interpret graph	Work inside out and track input/output.
Exponential Growth	Repeated percent change over time	Find the start a and the factor b .
Ratios, Rates & Conversion	“Per,” ratios, scale, mixed units	Write labeled fractions; cancel units.
Percentages	%, discount, increase, decrease	Turn the change into one multiplier.
One-Variable Data	Mean, median, spread, outliers	Decide: center or spread?
Scatterplots & Models	Trend, slope, line of best fit	Interpret slope with units.
Conditional Probability	“Given that” restricts the group	Shrink the denominator first.
Data Inferences & Claims	Sample, margin of error, conclusions	Identify the sampled population.

Area and Volume	Area, volume, or scaling a figure	Write the formula before substituting.
Lines, Angles & Triangles	Parallel lines, similar triangles	Mark every angle you can find.
Right Triangles	Missing side or trig ratio	Label opposite/adjacent/hypotenuse.
Circles	Arc, sector, equation, unit circle	Identify center, radius, and proportion.

ALGEBRA

Linear Equations in One Variable

Khan Academy: [Solving Linear Equations and Inequalities](#) · [Linear Equation Word Problems](#)

What this skill tests

Solve one-variable linear equations, including those arising from word problems, and recognize equations with no solution or infinitely many solutions.

How to recognize it

One variable, no exponents — or a word problem asking for a single unknown value.

Core strategy

Undo operations in reverse order, keeping both sides balanced.

Step-by-step method

1. Define the variable and its units.
2. Clear parentheses and fractions.
3. Collect variable terms on one side, constants on the other.
4. Divide by the coefficient of the variable.
5. Substitute back into the original equation to check.

Common traps

- Sign errors when distributing a negative.
- Solving for x but reporting the wrong quantity when the question asks for $2x + 1$.
- When the variable cancels entirely, the answer is no solution or infinitely many — not zero.

Mini-example

$$3(x - 2) = x + 8 \rightarrow 3x - 6 = x + 8 \rightarrow 2x = 14 \rightarrow x = 7.$$

Check your understanding

1. What does it mean if both sides reduce to the identical expression?
2. Why re-read what quantity the question actually asks for?

Linear Functions and Graphs

Khan Academy: [Graphs of Linear Equations and Functions](#) · [Linear Relationship Word Problems](#)

What this skill tests

Interpret slope and intercepts in context, and build the equation of a line from points, tables, graphs, or verbal descriptions.

How to recognize it

A constant rate of change; slope, intercept, or “per” language; a table with equal steps.

Core strategy

Read the slope as a rate with units and the y-intercept as the starting value.

Step-by-step method

1. Find two clean points, or identify the given rate.
2. Compute slope as change in y over change in x .
3. Identify the y-intercept — the value when $x = 0$.
4. Write $y = mx + b$.
5. Interpret both numbers with units in the context.

Common traps

- Swapping rise and run.
- Reading the x-intercept as the starting value.
- Assuming a table is linear without checking for equal differences.

Mini-example

$C = 15w + 40$ means \$15 per week plus a one-time \$40 fee; 40 is the cost at $w = 0$.

Check your understanding

1. What does the y-intercept mean in a word problem?
2. How do you confirm a table represents a linear function?

Solving Systems of Linear Equations

Khan Academy: [Solving Systems of Linear Equations](#) · [Systems of Linear Equations Word Problems](#)

What this skill tests

Find the ordered pair that satisfies two linear equations simultaneously, and recognize when a system has one, none, or infinitely many solutions.

How to recognize it

Two equations share x and y, or the question asks where two lines intersect.

Core strategy

Select substitution when a variable is isolated; select elimination when terms cancel easily.

Step-by-step method

1. Write both equations clearly and align like terms.
2. Choose substitution or elimination.
3. Reduce the system to one equation in one variable.
4. Solve, then substitute back to find the second variable.
5. Check the ordered pair in both original equations.

Common traps

- Substituting into the same equation instead of the other one.
- Sign errors during elimination.
- Reporting only x or only y.
- Missing no-solution or infinite-solution cases.

Mini-example

For $y = 2x + 1$ and $y = -x + 7$, set the expressions equal: $2x + 1 = -x + 7$. Then $x = 2$ and $y = 5$.

Check your understanding

1. When is elimination usually faster?
2. How do slopes reveal the number of solutions?

Linear Inequalities

Khan Academy: [Linear Inequality Word Problems](#) · [Graphs of Linear Systems and Inequalities](#)

What this skill tests

Solve and interpret one-variable or multi-constraint inequalities. Unlike equations, inequalities usually represent a range of valid values.

How to recognize it

Language includes more than, fewer than, at least, at most, no more than, or constraints on possible values.

Core strategy

Translate each condition separately and remember to reverse the sign when multiplying or dividing by a negative.

Step-by-step method

1. Define the variable and units.
2. Translate each verbal constraint into an inequality.
3. Simplify each side and isolate the variable.
4. Reverse the inequality if dividing or multiplying by a negative.
5. Apply real-world restrictions and test the final range.

Common traps

- Forgetting the sign reversal.
- Using the wrong symbol for at least or at most.
- Rounding in a direction that violates the constraint.
- Checking only one inequality in a system.

Mini-example

$-2x > 6$ becomes $x < -3$ after division by -2 . The direction reverses.

Check your understanding

1. What does “at most 12” mean symbolically?
2. Why can ordinary rounding give an invalid answer?

ADVANCED MATH

Equivalent Expressions

Khan Academy: [Operations with Polynomials](#) · [Factoring Quadratic and Polynomial Expressions](#) · [Radicals and Rational Exponents](#)

What this skill tests

Rewrite expressions — expand, factor, combine like terms, and apply exponent and radical rules — so that two forms can be matched.

How to recognize it

The stem says “equivalent to” or “which expression,” and asks for rewriting rather than solving.

Core strategy

Transform the given expression step by step toward the answer form, or plug in a test number to eliminate choices.

Step-by-step method

1. Scan the choices to see the target form.
2. Expand or factor deliberately, one step at a time.
3. Apply exponent rules: add exponents when multiplying, multiply when raising a power to a power.
4. Convert radicals to fractional exponents when helpful.
5. Verify with a test value such as $x = 2$.

Common traps

- $(a + b)^2$ is not $a^2 + b^2$.
- Adding exponents across unlike bases.
- $x^{1/2}$ is $\sqrt{x}$, not $x \div 2$.

Mini-example

$x^2 - 9$ factors as $(x - 3)(x + 3)$; testing $x = 4$ gives 7 for both forms.

Check your understanding

1. Why is plugging in a number a valid check for equivalence?
2. What is $x^{3/2}$ in radical form?

Isolating Quantities

Khan Academy: [Isolating Quantities](#)

What this skill tests

Rearrange a formula to express one variable in terms of the others. No numeric answer is required.

How to recognize it

A formula with several letters and a stem asking which equation gives one variable “in terms of” the rest.

Core strategy

Treat every other letter as a constant and undo operations in reverse order.

Step-by-step method

1. Mark the target variable.
2. Undo addition and subtraction first.
3. Undo multiplication and division next.
4. If the target sits in a denominator, multiply both sides by that denominator first.
5. Confirm the target appears exactly once, alone on one side.

Common traps

- Dividing only one term by a factor instead of the entire side.
- Mishandling a target variable that starts in the denominator.
- Sign errors when moving terms across the equals sign.

Mini-example

$d = rt$ rearranges to $t = d/r$.

Check your understanding

1. What extra first step is needed when the target variable is in a denominator?
2. Why can the other letters be treated as constants?

Solving Quadratic Equations

Khan Academy: [Solving Quadratic Equations](#) · [Quadratic Graphs](#)

What this skill tests

Solve equations of the form $ax^2 + bx + c = 0$ and determine the number of real solutions.

How to recognize it

The equation contains x^2 , or the question asks for roots, zeros, x-intercepts, or solution count.

Core strategy

Factor when it is quick; otherwise use square roots, completing the square, Desmos, or the quadratic formula.

Step-by-step method

1. Move all terms to one side so the equation equals zero.
2. Look for a common factor or easy factorization.
3. If factoring is not efficient, use the quadratic formula.
4. Include both roots when applicable.
5. Use the discriminant or graph to check the number of real solutions.

Common traps

- Forgetting the $\pm$ when taking square roots.
- Using incorrect a, b, or c signs.
- Stopping before applying the zero-product property.
- Assuming a negative discriminant has real roots.

Mini-example

$x^2 - 7x + 12 = 0$ factors as $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.

Check your understanding

1. What does a zero discriminant mean?
2. When is taking square roots the most direct method?

Linear and Quadratic Systems

Khan Academy: [Linear and Quadratic Systems](#)

What this skill tests

Find intersection points between a line and a parabola. Depending on the geometry, there may be zero, one, or two real solutions.

How to recognize it

One equation is linear and the other contains a squared term, or a graph shows a line with a parabola.

Core strategy

Substitute the linear expression into the quadratic equation and solve the resulting quadratic.

Step-by-step method

1. Isolate one variable in the linear equation.
2. Substitute that expression into the quadratic.
3. Simplify to standard quadratic form.
4. Solve for each possible x-value.
5. Back-substitute to find the matching y-values and check.

Common traps

- Keeping only one of two valid roots.
- Pairing an x-value with the wrong y-value.
- Assuming every line and parabola intersect twice.

Mini-example

If $y = x + 1$ and $y = x^2 - 3$, set $x + 1 = x^2 - 3$, solve the quadratic, then find y for each root.

Check your understanding

1. How does the graph show the number of solutions?
2. Why can one x-value produce a separate ordered pair?

Radical, Rational, and Absolute Value Equations

Khan Academy: [Radical, Rational, and Absolute Value Equations](#) • [Operations with Rational Expressions](#)

What this skill tests

Solve equations containing square roots, variables in denominators, or absolute values — and reject extraneous solutions.

How to recognize it

A radical sign, a variable in a denominator, or absolute-value bars in the equation.

Core strategy

Isolate the radical or absolute-value part first, and check every candidate in the original equation.

Step-by-step method

1. Isolate the radical, rational, or absolute-value expression.
2. Square both sides (radical), multiply through by the denominator (rational), or split into two cases (absolute value).
3. Solve the resulting equation.
4. Substitute each answer into the original equation.
5. Discard any value that fails the check or makes a denominator zero.

Common traps

- Keeping extraneous roots created by squaring.
- Forgetting the negative case of $|x| = k$.
- Multiplying both sides by an expression that could equal zero.

Mini-example

$\sqrt{x + 3} = x - 3$ leads to $x = 1$ or $x = 6$; only $x = 6$ survives the check.

Check your understanding

1. Why does squaring both sides create extraneous solutions?
2. How many cases does $|2x - 1| = 7$ create, and what are they?

Nonlinear Functions

Khan Academy: [Nonlinear Functions](#) · [Polynomial and Other Nonlinear Graphs](#)

What this skill tests

Evaluate, compose, and interpret polynomial, rational, and other nonlinear functions, including zeros, intercepts, end behavior, and restrictions.

How to recognize it

The problem uses $f(x)$, $f(g(x))$, a nonlinear graph, factored polynomial, or denominator restriction.

Core strategy

Track inputs and outputs carefully; for composition, work from the inside out.

Step-by-step method

1. Identify the requested input or feature.
2. Substitute the input for every x .
3. For $f(g(x))$, evaluate or substitute the inner function first.
4. Use factors to find zeros and the constant term for the y -intercept when appropriate.
5. Check domain restrictions, especially denominator $\neq 0$.

Common traps

- Treating $f(x)$ as multiplication.
- Evaluating the outer function first in a composition.
- Confusing zeros with the y -intercept.
- Including a value that makes a denominator zero.

Mini-example

If $f(x) = x^2 + 1$ and $g(x) = 2x$, then $f(g(3)) = f(6) = 37$.

Check your understanding

1. What is evaluated first in $f(g(x))$?
2. How are zeros related to x -intercepts?

Exponential Functions and Growth

Khan Academy: [Exponential Graphs](#) · [Quadratic and Exponential Word Problems](#) · [Linear and Exponential Growth](#)

What this skill tests

Build and interpret models of the form $y = a \cdot b^x$, distinguish exponential from linear growth, and read growth or decay factors.

How to recognize it

Repeated percent change over time, doubling or half-life, “increases by 4% per year,” or a curve changing by a constant factor.

Core strategy

Linear change adds a constant amount; exponential change multiplies by a constant factor. Identify the start a and the factor b .

Step-by-step method

1. Decide linear vs. exponential from the change pattern.
2. Identify the initial value a .
3. Convert percent change to a factor: 4% growth means $b = 1.04$.
4. Match the exponent to the stated time unit.
5. Interpret a and b in the context of the problem.

Common traps

- Writing $b = 0.04$ for 4% growth instead of 1.04.
- Mismatched time units in the exponent (per year vs. per month).
- Calling equal-difference growth exponential.

Mini-example

A population of 500 growing 3% per year: $P = 500(1.03)^t$.

Check your understanding

1. What factor represents a 12% decrease?
2. How does a table reveal exponential rather than linear change?

PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, Rates, Proportions, and Unit Conversion

Khan Academy: [Ratios, Rates, and Proportions](#) · [Unit Conversion](#)

What this skill tests

Set up and solve proportions, compute unit rates, and convert measurements between units.

How to recognize it

“Per,” “for every,” ratio notation, scale drawings, or mixed units within one problem.

Core strategy

Write every rate as a labeled fraction and let the units cancel.

Step-by-step method

1. Write the given rate with units in the numerator and denominator.
2. Set up the proportion so matching units align.
3. Chain conversion factors so unwanted units cancel.
4. Cross-multiply and solve.
5. Sanity-check the magnitude and units of the answer.

Common traps

- Flipping one ratio in a proportion but not the other.
- Converting only one dimension when units are squared or cubed.
- Confusing a ratio part with the total: 2:3 means the first part is $\frac{2}{5}$ of the whole.

Mini-example

180 miles in 3 hours is 60 mph; at that rate, 5 hours covers 300 miles.

Check your understanding

1. Why write units inside every fraction?
2. In a 3:4 ratio, what fraction of the total is the first part?

Percentages

Khan Academy: [Percentages](#)

What this skill tests

Find a percent of a quantity, compute percent change, and recover an original value before a change.

How to recognize it

A percent sign, “discount,” “increase,” “decrease,” or “what percent of.”

Core strategy

Translate to multiplication: $p\%$ of x is $(p/100) \cdot x$, and any change becomes a single multiplier.

Step-by-step method

1. Identify the base the percent acts on.
2. Convert the percent to a decimal.
3. Express a change as one multiplier: 20% off means $\times 0.80$.
4. To recover an original value, divide by the multiplier.
5. For percent change, compute $\text{change} \div \text{original}$.

Common traps

- A 20% increase followed by a 20% decrease does not return to the start.

- Percent change always uses the original value as the denominator.
- Applying a percent to the wrong base in a two-step problem.

Mini-example

A price is \$48 after 20% off, so the original is $48 \div 0.80 = \$60$.

Check your understanding

1. Why divide — not add the percent back — to find the original price?
2. What single multiplier represents a 15% increase?

One-Variable Data: Center, Spread, and Shape

Khan Academy: [Center, Spread, and Shape of Distributions](#) · [Data Representations](#)

What this skill tests

Work with mean, median, mode, range, and standard deviation, judge the effect of outliers, and read histograms, dot plots, and box plots.

How to recognize it

A data list or frequency display with a question about the typical value or how spread out the data are.

Core strategy

Decide whether the question is about center or spread, then choose the outlier-resistant measure when outliers exist.

Step-by-step method

1. Read the display and count the number of data points.
2. For the median, find the middle position — not the middle of the axis.
3. For a mean from a frequency table, use $\text{value} \times \text{frequency}$.
4. Compare two distributions by both center and spread.
5. Ask how an outlier moves the mean versus the median.

Common traps

- An outlier pulls the mean strongly but barely moves the median.
- A tighter cluster means a smaller standard deviation — no calculation needed.
- Reading the midpoint of the axis as the median.

Mini-example

For 2, 3, 3, 4, 50 the median 3 describes the typical value; the mean 12.4 does not.

Check your understanding

1. Which is more resistant to outliers, the mean or the median?
2. What does a larger standard deviation indicate?

Two-Variable Data: Models and Scatterplots

Khan Academy: [Scatterplots](#)

What this skill tests

Interpret a scatterplot, line of best fit, slope, intercept, residual, and the appropriateness of linear or nonlinear models.

How to recognize it

A graph contains many data points and a trend line, or the question asks for a prediction or interpretation of slope.

Core strategy

Use the line of best fit for the overall trend and interpret every coefficient with units.

Step-by-step method

1. Read axes, units, and the plotted range.
2. Identify direction, form, and strength of association.
3. Estimate slope as change in y per one unit of x .
4. Use the model — not an individual point — for prediction.
5. Avoid strong extrapolation far outside the observed range.

Common traps

- Reading one data point as the trend.
- Describing correlation as causation.
- Ignoring units in slope.
- Confusing constant change with constant percentage change.

Mini-example

A slope of 2.5 dollars per item means the model predicts cost rises by \$2.50 for each additional item.

Check your understanding

1. What does a residual measure?
2. Why is extrapolation less reliable?

Probability and Conditional Probability

Khan Academy: [Probability and Relative Frequency](#)

What this skill tests

Calculate probabilities from counts, two-way tables, or relative frequencies. Conditional probability restricts the sample space to a specified subgroup.

How to recognize it

The question includes “given that,” “among,” “of those,” or a two-way frequency table.

Core strategy

Determine the correct denominator before calculating.

Step-by-step method

1. Add missing row and column totals if necessary.
2. Identify the favorable outcomes.
3. Identify the allowed population or subgroup.
4. Compute favorable $\div$ allowed total.
5. Check that the probability is between 0 and 1.

Common traps

- Using the grand total for a conditional question.
- Reversing the condition and the event.
- Using percentages and counts inconsistently.

Mini-example

If 18 of 30 students in the Grade 11 subgroup chose math, then $P(\text{math} \mid \text{Grade 11}) = 18/30$, not 18 divided by the entire program.

Check your understanding

1. What does “given that” do to the denominator?
2. How can you distinguish $P(A \mid B)$ from $P(B \mid A)$?

Data Inferences and Statistical Claims

Khan Academy: [Data Inferences](#) · [Evaluating Statistical Claims](#)

What this skill tests

Generalize from a sample using proportions and margin of error, and judge which conclusions a study design actually supports.

How to recognize it

“Randomly selected,” “margin of error,” or “which conclusion is best supported.”

Core strategy

Random sampling allows generalizing to the sampled population; only random assignment allows cause-and-effect claims.

Step-by-step method

1. Identify the population that was actually sampled.
2. Scale the sample proportion up to that population.
3. Read margin of error as an interval around the estimate.
4. Check the design: random selection? random assignment?
5. Reject any conclusion broader than the design supports.

Common traps

- Drawing a causal claim from observational data.
- Generalizing beyond the population that was sampled.
- Treating the estimate as exact when a margin of error is given.

Mini-example

If 60% of 200 randomly sampled students agree, with $\pm 5\%$ margin of error, the plausible school-wide rate is 55–65% — not exactly 60%.

Check your understanding

1. What does the margin of error tell you about the estimate?
2. Which design feature justifies a cause-and-effect conclusion?

GEOMETRY AND TRIGONOMETRY

Area and Volume

Khan Academy: [Area and Volume](#)

What this skill tests

Find area and perimeter of standard shapes, volume and surface area of solids, and the effect of scaling dimensions.

How to recognize it

A figure with dimensions, or the words “area,” “volume,” or “how many times larger.”

Core strategy

Write the correct formula (the test's reference sheet has them) before substituting, and keep units consistent.

Step-by-step method

1. Identify the shape or solid.
2. Write the formula before substituting numbers.
3. Convert all lengths to the same unit.
4. Substitute and compute.
5. For scaling: area scales by k^2 , volume by k^3 .

Common traps

- Using the diameter where the formula requires the radius.
- Doubling a side doubles the area — it actually quadruples it.
- Mishandling composite figures when subtracting regions.

Mini-example

Doubling a cube's edge multiplies its volume by 8.

Check your understanding

1. If every side of a figure triples, what happens to its area?
2. Why write the formula before substituting?

Lines, Angles, and Triangles

Khan Academy: [Congruence, Similarity, and Angle Relationships](#)

What this skill tests

Use angle relationships from parallel lines, the triangle angle sum, and congruence and similarity to find unknown angles and sides.

How to recognize it

Parallel lines cut by a transversal, similar triangles, or unknown angles in a figure.

Core strategy

Mark every angle you can determine, then use similarity ratios for unknown sides.

Step-by-step method

1. Label all given angles and sides on the figure.
2. Fill in angles using vertical, supplementary, and parallel-line pairs.
3. Apply the 180° triangle angle sum.
4. For similar triangles, match corresponding vertices carefully.
5. Set up the side proportion and solve.

Common traps

- Pairing non-corresponding sides in a similarity proportion.
- Assuming similarity without two equal angles.
- An exterior angle equals the sum of the two remote interior angles — not the adjacent one.

Mini-example

If two triangles share two angle measures, they are similar, so all side pairs scale by one ratio.

Check your understanding

1. What is the quickest way to confirm two triangles are similar?
2. What does an exterior angle of a triangle equal?

Right Triangles and Trigonometry

Khan Academy: [Right Triangle Trigonometry](#)

What this skill tests

Find unknown sides or angles using the Pythagorean theorem, special triangles, similarity, and sine, cosine, or tangent.

How to recognize it

The diagram contains a right angle, an acute angle with side lengths, or similar right triangles.

Core strategy

Label sides relative to the given angle before selecting a trig ratio.

Step-by-step method

1. Mark the hypotenuse first.
2. Relative to the angle, label opposite and adjacent.
3. Choose SOH, CAH, or TOA — or use a special triangle ratio.
4. Solve algebraically and maintain exact values when requested.
5. Check that the answer is reasonable for the diagram.

Common traps

- Using the wrong reference angle.
- Calling a leg the hypotenuse.
- Mixing degrees and radians.
- Forgetting both legs in the Pythagorean theorem.

Mini-example

In a 3-4-5 triangle, if the side opposite θ is 3 and the hypotenuse is 5, then $\sin \theta = 3/5$.

Check your understanding

1. Which side is always the hypotenuse?
2. How do 30-60-90 side ratios differ from 45-45-90?

Circles

Khan Academy: [Circle Theorems](#) • [Circle Equations](#) • [Unit Circle Trigonometry](#)

What this skill tests

Solve problems involving arc length, sector area, central angles, circle equations, radii, chords, and unit-circle coordinates.

How to recognize it

The problem includes a circle diagram, an equation with squared x and y terms, or an angle measured around the unit circle.

Core strategy

Identify the center and radius first, then use proportional or coordinate relationships.

Step-by-step method

1. For diagrams, mark all radii as equal.
2. Relate central angle/360 to arc/circumference or sector/circle area.
3. For equations, convert to $(x-h)^2 + (y-k)^2 = r^2$ when needed.
4. For unit-circle questions, reduce angles by full rotations.
5. Read cosine as the x-coordinate and sine as the y-coordinate.

Common traps

- Reading the center signs incorrectly.
- Using diameter where the formula requires radius.
- Confusing arc length with sector area.
- Failing to reduce a large angle modulo 2π .

Mini-example

In $(x-3)^2 + (y+2)^2 = 25$, the center is $(3, -2)$ and the radius is 5.

Check your understanding

1. Why is the center's y-coordinate -2 in the example?
2. How are central angle, arc length, and sector area proportionally related?